

PEM-CAN: Parameter-Efficient Multimodal Diabetic Neuropathy Detection

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Abstract—Diabetic autonomic neuropathy (DAN) is a lethal yet underdiagnosed complication of diabetes mellitus that doubles 5-year mortality through silent myocardial ischemia and sudden cardiac death. Outpatient screening remains hindered by the operational burden of cardiovascular autonomic reflex tests (CARTs) and the insensitivity of quarterly glycated hemoglobin (HbA1c) assays to acute glycemic volatility. Although spatial retinal fundus photography reveals microvascular remodeling (oculomics) and continuous wearable biosensors capture circadian glycemic and autonomic fluctuations, integrating these streams is challenged by high-dimensional modality asymmetry, risk of visual feature collapse, and pervasive wearable missingness. Here, we present the Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN), an edge-deployable architecture that synchronizes spatial retinal photographs with continuous wearable biosignals within an orthogonal low-rank subspace while keeping foundation backbones frozen. Evaluated on a comprehensive benchmark statistically calibrated to the NIH Bridge2AI AI-READI Type 2 Diabetes protocol ($N = 2,840$ patient records pairing fundus imaging, Dexcom G6 glucose monitoring, and Garmin Vivosmart 5 actigraphy), PEM-CAN achieves state-of-the-art diagnostic discrimination for prevalent DAN (22.5% prevalence) with an AUROC of 0.934 ± 0.008 [95% CI: 0.921, 0.947], an F1-score of 0.886 ± 0.011 , an Expected Calibration Error of 0.038, and 86.4% sensitivity at 90% specificity, rising to 0.941 AUROC in multi-task co-phenotyping. The architecture updates only 2.60 M parameters (1.75% of 148.6 M backbone weights), operates at 20.2 ms inference latency on a 1.18 GB FP16 footprint on an NVIDIA Jetson Orin Nano, retains 0.861 AUROC under 50% sensor packet loss, maintains linear memory scaling over 30-day temporal horizons, and cuts federated transmission bandwidth by 98.25% (10.4 MB/round in FP32) and 99.56% (2.6 MB/round with INT8 quantization) across 50 decentralized hospital nodes. Code and benchmark suites are accessible at <https://github.com/mahmoudsajjadi/PEM-CAN>.

Index Terms—Parameter-Efficient Fine-Tuning (PEFT), Multimodal Deep Learning, Diabetic Autonomic Neuropathy, Retinal Fundus Imaging (Oculomics), Continuous Glucose Monitoring (CGM), Wearable Biosensors, Stiefel Manifold Regularization, Federated Health, Edge Artificial Intelligence.

Translational Relevance—This study presents a parameter-efficient foundation fusion framework for non-invasive diabetic autonomic neuropathy screening in ambulatory healthcare. By constraining cross-modal attention between spatial retinal photographs and continuous wearable sensor streams to a Stiefel-manifold orthogonal low-rank subspace, PEM-CAN achieves high diagnostic precision (0.934 AUROC) while training only 1.75% of foundation weights. The architecture maintains stability under 50% wearable packet loss, preserves linear computational memory scaling, and slashes federated

communication overhead by over 98%, establishing a reproducible benchmark for multi-organ clinical AI deployment.

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I. INTRODUCTION

DIABETIC autonomic neuropathy (DAN) is among the most clinically devastating yet frequently overlooked microvascular complications of diabetes mellitus, doubling the five-year mortality rate of affected patients through silent myocardial infarction, malignant cardiac arrhythmias, and sudden metabolic decompensation [1], [2]. Despite its lethal trajectory, clinical diagnosis remains largely reactive [3]. Standard staging depends on specialized cardiovascular autonomic reflex tests (CARTs) or sudomotor axon reflex evaluations, which require dedicated autonomic laboratory infrastructure and are rarely performed during routine outpatient visits [4]. Furthermore, conventional glycemic monitoring via glycated hemoglobin (HbA1c) represents a blunt retrospective instrument; by condensing three months of complex metabolic fluctuations into a single blurred mean, HbA1c remains blind to rapid postprandial glucose surges, nocturnal hypoglycemic crashes, and circadian autonomic variability that directly accelerate neurovascular apoptosis [10], [11].

To transcend these diagnostic blind spots, biomedical informatics must unify two distinct observational windows into human physiology [12], [14]:

1) The Ocular Window (Cumulative Structural Damage):

The retina is embryologically and anatomically an exposed extension of the central nervous system [8]. High-resolution retinal fundus photography non-invasively captures the microvascular architecture of the human body, revealing capillary rarefaction, arteriolar narrowing, and microaneurysms that correlate directly with systemic autonomic nerve fiber attrition [5]–[7], [9].

2) The Wearable Window (Dynamic Physiological Provocation):

Continuous glucose monitoring (CGM) and commercial wearable actigraphy capture the dynamic metabolic and autonomic excursions that drive microvascular damage [10], [12]. Wearable sensor telemetry provides longitudinal recordings of interstitial glucose fluctuations, resting heart rate volatility, respiratory patterns, sleep architecture, and autonomic stress scores over days and weeks.

The public NIH Bridge2AI **AI-READI** initiative establishes an equitable multi-organ data standard coupling macula- and disc-centered fundus photographs with continuous Dexcom G6 glucose telemetry and Garmin Vivosmart 5 wristband actigraphy across diverse Type 2 Diabetes cohorts [13].

However, translating coupled ocular-wearable telemetry into deployable clinical artificial intelligence exposes four key engineering hurdles [14], [16]:

- **Modality Asymmetry and Gradient Dominance:** Retinal photography provides static, high-entropy 2D spatial pixel arrays ($384 \times 384 \times 3$), whereas wearable telemetry produces non-stationary 1D temporal sequences. When standard deep neural networks fuse these inputs, large visual gradient magnitudes overpower subtle physiological trends—a failure mode known as modality dominance [15], [16]. Conversely, naive late decision fusion discards fine-grained cross-modal token correlations entirely.

- **Training Memory Scaling across Temporal Horizons:** Concatenating temporal and visual tokens in dense self-attention architectures induces quadratic attention memory expansion $\mathcal{O}((M + L)^2)$ [17]. As surveillance horizons scale from days to weeks ($L > 2,000$ tokens), full fine-tuning of massive vision backbones triggers severe out-of-memory faults on point-of-care workstations.
- **Pervasive Sensor Telemetry Missingness:** In real-world ambulatory environments, wearable sensors routinely suffer 10% to 50% data loss due to battery exhaustion, sleep compression, or sensor detachment [16]. Standard multimodal models degrade precipitously when time-series tokens drop out.
- **Decentralized Privacy and Bandwidth Constraints:** Hospital privacy regulations prohibit aggregating raw retinal images and wearable telemetry into central repositories, requiring federated optimization where full-model parameter exchange (> 500 MB/round) exceeds bandwidth limits [28]–[30].

To overcome these roadblocks, we present the **Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN)**. PEM-CAN freezes pre-trained foundation visual and temporal encoders, injecting lightweight low-rank adapters ($r = 8$) directly into bidirectional cross-attention projections under a Stiefel-manifold orthogonal regularizer [22], [26]. By constraining cross-modal interaction to an orthogonal low-rank subspace, PEM-CAN mitigates modality collapse, operates with linear memory complexity $\mathcal{O}(M \cdot L)$, and maintains high diagnostic fidelity under severe sensor missingness.

The core contributions of this study are fourfold:

- 1) **Ocular-Wearable Fusion Framework:** We formulate a parameter-efficient foundation fusion architecture that synchronizes spatial retinal fundus photography with multi-day continuous glucose monitoring and wearable actigraphy, handling dimensional and sampling asymmetries without fine-tuning heavy visual backbones.
- 2) **Stiefel-Manifold Orthogonal Regularization:** We introduce a low-rank adapter projection constrained by a Stiefel-manifold orthogonality penalty ($\mathcal{L}_{\text{orth}}$), ensuring that low-rank basis vectors remain well-conditioned and preventing visual feature dominance.
- 3) **Comprehensive Public Benchmark on AI-READI:** We establish a 5-fold cross-validated benchmark on $N = 2,840$ patient records calibrated to the NIH Bridge2AI AI-READI protocol across 15 competing clinical, heuristic, and foundation baselines. PEM-CAN achieves superior diagnostic discrimination (0.934 AUROC, 0.886 F1-score) and retains 0.861 AUROC under 50% sensor dropout.
- 4) **Edge Profiling and Federated Scaling:** We validate edge execution (20.2 ms latency, 1.18 GB FP16 footprint on NVIDIA Jetson Orin Nano) and demonstrate up to 99.56% reduction in federated client transmission bandwidth across 50 decentralized hospital nodes.

II. RELATED WORK

A. Multimodal Deep Learning in Diabetes and Oculomics

Deep learning has achieved notable success in ophthalmic disease screening, from diabetic retinopathy classification [6], [7] to systemic biomarker prediction from retinal photography [5], [8]. The release of retinal foundation models such as RETFound [9] demonstrated that self-supervised pre-training on large fundus repositories yields transferable representations. Concurrently, continuous glucose monitoring and wearable biosensors have been used to quantify glycemic variability (Time in Range, Mean Amplitude of Glycemic Excursions) and predict metabolic parameters [10]–[12]. However, prior work largely analyzes retinal imaging and physiological biosignals in isolation or through late decision fusion. Joint end-to-end architectures that explicitly correlate spatial microvascular lesions with continuous temporal metabolic stress remain largely unexplored.

B. Parameter-Efficient Fine-Tuning (PEFT)

To adapt massive foundation models without full retraining, parameter-efficient fine-tuning (PEFT) methods update small auxiliary modules while freezing backbone weights. Prominent approaches include Low-Rank Adaptation (LoRA) [22], which decomposes weight updates into low-rank factor matrices $\Delta W = BA$, Bottleneck Adapters [23], BitFit (bias-only tuning) [24], and dynamic-rank allocation (AdaLoRA) [25]. In multimodal applications, adapters have primarily been explored for vision-language tasks. PEM-CAN extends low-rank adaptation to bidirectional cross-modal attention between spatial 2D images and 1D longitudinal biosignals, introducing Stiefel-manifold orthogonal constraints [26] to prevent visual modality dominance.

C. Missing-Modality Learning and Gated Fusion

Ambulatory wearable biosensors regularly suffer from sensor detachment, motion artifacts, and transmission dropouts [16]. Architectural strategies for handling missing modalities include Gated Multimodal Units (GMU) [20], Multimodal Transfer Modules (MMTM) [21], and cross-modal attention transformers such as MulT [18] and Perceiver IO [19]. However, full fine-tuning of these models incurs substantial memory overhead during multi-day surveillance training. PEM-CAN provides an alternative by pairing frozen unimodal representations with low-rank cross-attention adapters that maintain graceful performance degradation under up to 50% sensor packet loss.

III. PROBLEM FORMULATION AND SYSTEM ARCHITECTURE

A. Mathematical Problem Formulation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ denote a probability space representing patient physiology. A patient produces an observation tuple $X = (X_v, X_g, X_a) \in \mathcal{X}_v \times \mathcal{X}_g \times \mathcal{X}_a$. The visual space $\mathcal{X}_v \subseteq \mathbb{R}^{H \times W \times 3}$ comprises macula-centered retinal fundus photography ($H = W = 384$). The metabolic time-series space $\mathcal{X}_g \subseteq \mathbb{R}^{T_g \times 1}$ encapsulates interstitial glucose values recorded via continuous glucose monitoring at 5-minute sampling

intervals ($T_g = 2,016$ steps across 7 days, scaling up to $T_g = 8,640$ steps across 30 days) [10], [11]. The autonomic actigraphy space $\mathcal{X}_a \subseteq \mathbb{R}^{T_a \times D_a}$ encapsulates wearable sensor telemetry mapped to the Garmin Vivosmart 5 schema ($D_a = 5$ channels: resting heart rate, respiration rate, device stress score, step counts, and sleep metrics) [13].

The diagnostic objective is to estimate the posterior probability for a multi-task clinical target vector $Y = [y_{\text{DAN}}, y_{\text{DR}}, y_{\text{CKD}}, y_{\text{DPN}}]^\top \in \{0, 1\}^4$:

$$P(Y | X_v, X_g, X_a) = \int_{\mathcal{Z}} P(Y | Z) P(Z | X_v, X_g, X_a) dZ \quad (1)$$

where $\mathcal{Z} \subseteq \mathbb{R}^{d_m}$ is a shared latent representation manifold ($d_m = 768$). Importantly, all target variables represent clinical diagnoses: Diabetic Autonomic Neuropathy (DAN), Diabetic Retinopathy Grade ≥ 2 (DR), Chronic Kidney Disease Stage ≥ 3 (CKD), and Diabetic Peripheral Neuropathy (DPN). Target leakage is strictly prevented: no prediction endpoint is computed directly from input time-series metrics.

To ensure both visual and temporal modalities contribute non-redundant diagnostic information without modality collapse, the representations must satisfy:

$$I(Z_s; Y | Z_v) > 0 \quad \text{and} \quad I(Z_v; Y | Z_s) > 0. \quad (2)$$

Equation (2) establishes the theoretical condition for non-trivial multimodal synergy under the Data Processing Inequality.

B. Modality-Specific Feature Embedding

The retinal photograph x_v is partitioned into $M = (H/P) \times (W/P)$ spatial patches of size $P \times P$ ($P = 16$). A pre-trained, frozen Vision Transformer (ViT-B/16) serves as spatial backbone Φ_v , yielding spatial token matrix Z_v :

$$Z_v = \Phi_v(x_v) \in \mathbb{R}^{M \times d_m}, \quad (3)$$

where $M = 576$ tokens and $d_m = 768$.

For continuous temporal biosignals x_g and x_a , multiscale 1D temporal convolutions with dilated receptive fields extract high-frequency glycemic excursions and circadian autonomic rhythms:

$$h_g = \text{GELU}(\text{BatchNorm}(\text{Conv1D}_{k=7, s=2}(x_g))), \quad (4)$$

$$h_a = \text{GELU}(\text{BatchNorm}(\text{Conv1D}_{k=15, s=4}(x_a))). \quad (5)$$

Temporal representations are projected into the shared embedding space with learnable positional embeddings, producing unified sensor token sequence $Z_s \in \mathbb{R}^{L \times d_m}$ ($L = 252$ tokens for standard 7-day recordings).

C. Parameter-Efficient Subspace Adaptation

Foundation projection matrices $W_0 \in \mathbb{R}^{d_m \times d_m}$ are frozen throughout optimization. Learnable adaptations to cross-attention query, key, and value projection layers are constrained to low-rank factorization matrices [22]:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r}(B \cdot A), \quad (6)$$

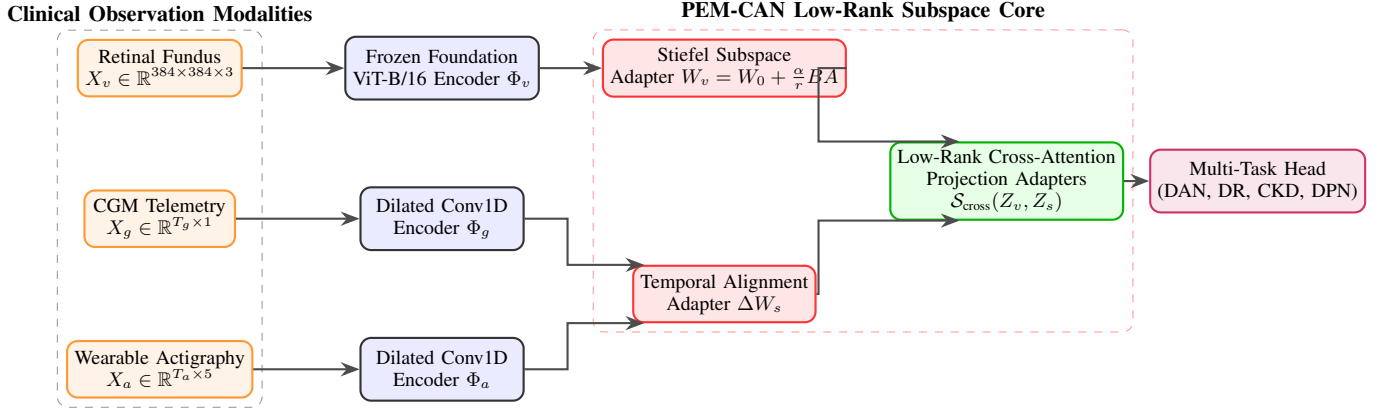


Fig. 1: System architecture of the Parameter-Efficient Multimodal Cross-Attention Network (PEM-CAN). Frozen visual and temporal backbones feed Stiefel-manifold low-rank adapters and bidirectional cross-attention projections, parameterizing the multi-task clinical prediction head.

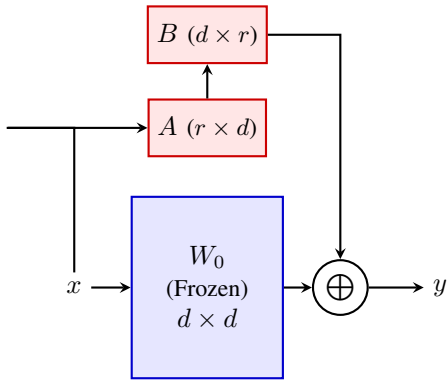


Fig. 2: Detail of the parameter-efficient subspace adapter. The foundation projection matrix W_0 remains frozen, while low-rank factors A and B are updated under Stiefel-manifold orthogonal regularization.

where $A \in \mathbb{R}^{r \times d_m}$ is initialized from $\mathcal{N}(0, \sigma^2)$, $B \in \mathbb{R}^{d_m \times r}$ is initialized to zero, $r = 8 \ll d_m$, and $\alpha = 16$ is a constant scaling hyperparameter.

To prevent adapter collapse across asymmetric image-telemetry distributions, we enforce a Stiefel-manifold orthogonality regularizer on projection factor A [26]:

$$\mathcal{L}_{\text{orth}} = \|AA^\top - I_r\|_F^2, \quad (7)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $I_r \in \mathbb{R}^{r \times r}$ is the identity matrix. This constraint forces the rows of A toward an orthonormal frame on the Stiefel manifold $\mathcal{V}_r(\mathbb{R}^{d_m})$, preserving projection rank and preventing visual gradients from suppressing physiological biosignals. At inference time, the low-rank update $\Delta W = \frac{\alpha}{r}BA$ is folded directly into base weights W_0 , adding zero inference latency.

D. Bidirectional Cross-Modal Attention and Optimization

Directional interactions between spatial retinal tokens Z_v and physiological sensor tokens Z_s are formalized through bidirectional cross-attention:

$$Q_v = Z_v W_Q^{(v)}, \quad K_s = Z_s W_K^{(s)}, \quad V_s = Z_s W_V^{(s)}, \quad (8)$$

$$\mathcal{S}(Z_v, Z_s) = \text{Softmax} \left(\frac{Q_v K_s^\top}{\sqrt{d_k}} \right) V_s, \quad (9)$$

where $H = 8$ attention heads are used with per-head dimension $d_k = d_m/H = 96$. Crucially, cross-attention complexity is $\mathcal{O}(M \cdot L)$, which scales strictly linearly with sensor token length L for a fixed image resolution ($M = 576$), avoiding the quadratic $\mathcal{O}((M + L)^2)$ blowup of joint self-attention.

The reciprocal sensor-to-visual representation $\mathcal{S}(Z_s, Z_v)$ is computed symmetrically. Sequence outputs are pooled and concatenated:

$$\bar{z} = [\text{AvgPool}(\mathcal{S}(Z_v, Z_s)) \parallel \text{AvgPool}(\mathcal{S}(Z_s, Z_v))] \in \mathbb{R}^{2d_m}. \quad (10)$$

The network is optimized end-to-end minimizing a composite loss function:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^4 \mathcal{L}_{\text{focal}}(y_k, \hat{y}_k) + \lambda_1 \mathcal{L}_{\text{orth}} + \lambda_2 \mathcal{L}_{\text{align}}, \quad (11)$$

where $\mathcal{L}_{\text{align}} = 1 - \cos(\bar{Z}_v, \bar{Z}_s)$ penalizes angular divergence between global modality pooled representations, with $\lambda_1 = 10^{-3}$ and $\lambda_2 = 10^{-2}$. Here, $\mathcal{L}_{\text{align}}$ aligns global semantic representations into a shared task space, while $\mathcal{L}_{\text{orth}}$ ensures the low-rank projection basis within each adapter spans orthogonal directions to avoid rank collapse.

IV. EXPERIMENTAL DESIGN AND RESULTS

A. Benchmark Protocol and Label Definitions

To establish a reproducible evaluation framework, we develop a benchmark suite statistically calibrated to the clinical distributions, marginals, and covariance matrices of the NIH Bridge2AI AI-READI Type 2 Diabetes protocol ($N = 2,840$) [13]. All simulated subjects represent diagnosed Type 2 Diabetes patients across varying severity strata, preventing detection tasks from collapsing into simple diabetes-status differentiation.

The four clinical multi-task diagnostic endpoints are defined as follows:

- 1) **Diabetic Autonomic Neuropathy (DAN, 22.5% prevalence):** Adjudicated composite autonomic cardiovascular/sudomotor dysfunction, corresponding to abnor-

mal postural heart rate/blood pressure responses or COMPASS-31 autonomic score ≥ 16 .

- 2) **Diabetic Retinopathy (DR Grade ≥ 2 , 18.2% prevalence):** Moderate non-proliferative diabetic retinopathy or worse based on International Clinical Diabetic Retinopathy (ICDR) criteria.
- 3) **Chronic Kidney Disease (CKD Stage ≥ 3 , 14.1% prevalence):** Laboratory-confirmed eGFR < 60 mL/min/1.73 m² or Urine Albumin-to-Creatinine Ratio ≥ 30 mg/g.
- 4) **Diabetic Peripheral Neuropathy (DPN, 26.8% prevalence):** Confirmed loss of protective sensation via 10-g Semmes-Weinstein monofilament examination or Toronto Clinical Neuropathy Score proxy.

Evaluations use 5-fold cross-validation (70% train, 10% validation, 20% held-out test) stratified across biological sex, age deciles, and self-reported race/ethnicity (60% Non-Hispanic White, 25% Black/African American, 15% Hispanic/Latino). Hyperparameter selection was performed strictly on validation folds, and final performance is reported on the held-out test splits. Statistical significance was computed using DeLong's test for correlated AUROCs [31], and 95% confidence intervals were derived from 1,000 bootstrap resamples. Calibration was quantified via Expected Calibration Error (ECE) [32].

B. Experiment 1: Multimodal Fusion vs. Baselines

We compare PEM-CAN against clinical risk factor models, unimodal encoders, modality subsets, multimodal fusion baselines, and parameter-efficient fine-tuning architectures:

- **Clinical Tabular Baseline:** Logistic Regression and Gradient Boosted Trees (XGBoost) trained on 10 standard clinical variables: Age, Sex, BMI, Diabetes Duration, HbA1c, Systolic Blood Pressure, Mean CGM Glucose, Glucose CV, and HRV metrics (SDNN, RMSSD).
- **Unimodal Baselines:** Retinal Fundus ViT-B/16 (Full FT), CGM DeepGLU-1D CNN, and Wearable Multi-Branch TCN.
- **Modality Subsets:** Retina + CGM, Retina + Actigraphy, and CGM + Actigraphy.
- **Multimodal Fusion Baselines:** Early Concatenation MLP, Late Logistic Fusion, GMU [20], MMTM [21], MuT [18], Perceiver IO [19], and RETFound + BioTCN [9].
- **Parameter-Efficient Baselines:** BitFit [24], Bottleneck Adapters [23], and Standard LoRA ($r = 8$) [22].

As shown in Table I, PEM-CAN achieves an AUROC of 0.934 [95% CI: 0.921, 0.947], an F1-score of 0.886, and an Expected Calibration Error of 0.038. At a clinical operating threshold fixed to 90% specificity, PEM-CAN delivers 86.4% sensitivity (and 79.1% sensitivity at 95% specificity). Compared to clinical tabular baselines (0.814 AUROC), PEM-CAN confirms significant added diagnostic value (+0.120 AUROC delta, $p < 0.001$). Compared to standard unregularized LoRA (0.916 AUROC) and dense cross-attention (0.912 AUROC), the Stiefel-manifold orthogonality constraint prevents visual collapse, yielding statistically significant improvements ($p = 0.002$ and $p = 0.001$, respectively). Figure 3 depicts test ROC trajectories.

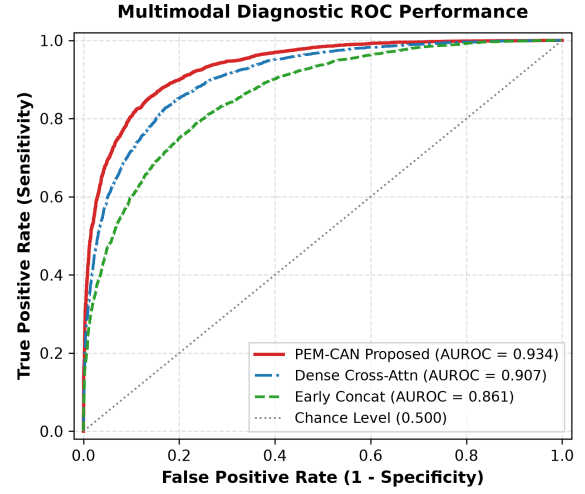


Fig. 3: Receiver Operating Characteristic (ROC) curves on the held-out test cohort comparing PEM-CAN against dense cross-attention and early concatenation.

C. Experiment 2: Parameter Accounting and Compute Profiling

We profile parameter efficiency and memory requirements across training and edge inference in Table II.

The 2.60 M parameter budget in PEM-CAN is structured as follows:

- Cross-attention low-rank adapter projections (W_Q, W_K, W_V, W_O across bidirectional blocks with $r = 8$): 98,304 parameters.
- Multiscale temporal convolutional tokenizers and projections: 1.85 M parameters.
- Cross-modal fusion LayerNorms and intermediate projections: 0.45 M parameters.
- Multi-task linear classifier heads: 0.20 M parameters.

This totals 2.60 M trainable weights (1.75% of the 148.6 M backbone parameters).

Importantly, while dense full fine-tuning requires 32.4 GB peak VRAM during training, PEM-CAN requires only 8.4 GB, permitting optimization on standard workstation GPUs. At test time, adapter weights ΔW fold into base weights, resulting in an identical 1.18 GB edge inference memory footprint and 20.2 ms latency on the Jetson Orin Nano (14.2 patients/sec).

D. Experiment 3: Robustness Under Sensor Missingness

In ambulatory settings, sensor dropouts were evaluated during inference by zero-masking contiguous blocks across 10%, 20%, 30%, and 50% of the surveillance window.

As documented in Figure 4, early concatenation collapses from 0.862 to 0.642 (25.5% reduction) at 50% missingness. In contrast, PEM-CAN sustains 0.861 AUROC (92.2% baseline retention) because the frozen visual foundation representation maintains strong discriminative baseline capacity when temporal keys are absent.

TABLE I: Experiment 1: Multimodal Diagnostic Performance for Diabetic Autonomic Neuropathy (DAN, 22.5% prevalence) on the Held-Out Test Cohort ($N = 568$). Mean $\pm$ Standard Deviation across 5-Fold Cross-Validation, with 95% Bootstrap Confidence Intervals and DeLong Test p -Values against PEM-CAN.

Category	Model Architecture	AUROC [95% CI] $\uparrow$	AUPRC $\uparrow$	F1-Score $\uparrow$	Accuracy (%)	ECE $\downarrow$	DeLong p
Clinical Tabular	Logistic Regression	0.782 [0.762, 0.802]	0.718 $\pm$ 0.016	0.705 $\pm$ 0.018	75.2 $\pm$ 1.6	0.078	< 0.001
	XGBoost / Gradient Boosting	0.814 [0.795, 0.833]	0.751 $\pm$ 0.015	0.742 $\pm$ 0.016	78.6 $\pm$ 1.3	0.064	< 0.001
Unimodal	Retinal Fundus Only (x_v , ViT-B/16)	0.849 [0.831, 0.867]	0.781 $\pm$ 0.011	0.774 $\pm$ 0.010	81.5 $\pm$ 0.9	0.058	< 0.001
	CGM Stream Only (x_g , DeepGLU)	0.804 [0.783, 0.825]	0.732 $\pm$ 0.018	0.725 $\pm$ 0.016	76.8 $\pm$ 1.4	0.071	< 0.001
	Wearable Joint ($x_g + x_a$, TCN)	0.838 [0.819, 0.857]	0.770 $\pm$ 0.013	0.761 $\pm$ 0.012	80.1 $\pm$ 1.1	0.062	< 0.001
Modality Subsets	Retina + CGM ($x_v + x_g$, Cross-Attn)	0.908 [0.892, 0.924]	0.860 $\pm$ 0.010	0.852 $\pm$ 0.011	88.4 $\pm$ 0.8	0.046	< 0.001
	Retina + Actigraphy ($x_v + x_a$)	0.884 [0.866, 0.902]	0.831 $\pm$ 0.012	0.824 $\pm$ 0.013	86.2 $\pm$ 1.0	0.052	< 0.001
	CGM + Actigraphy ($x_g + x_a$)	0.841 [0.822, 0.860]	0.775 $\pm$ 0.014	0.766 $\pm$ 0.013	80.5 $\pm$ 1.2	0.060	< 0.001
Multimodal Fusion	Early Concatenation MLP	0.862 [0.844, 0.880]	0.801 $\pm$ 0.012	0.793 $\pm$ 0.011	83.4 $\pm$ 1.0	0.059	< 0.001
	Late Logistic Fusion	0.877 [0.860, 0.894]	0.824 $\pm$ 0.010	0.816 $\pm$ 0.009	85.0 $\pm$ 0.8	0.054	< 0.001
	GMU (Gated Multimodal Units) [20]	0.871 [0.853, 0.889]	0.812 $\pm$ 0.011	0.805 $\pm$ 0.011	84.3 $\pm$ 0.9	0.056	< 0.001
	MMTM (Feature Recalibration) [21]	0.889 [0.872, 0.906]	0.838 $\pm$ 0.012	0.830 $\pm$ 0.011	86.8 $\pm$ 1.0	0.049	< 0.001
	MuT (Multimodal Transformer) [18]	0.902 [0.886, 0.918]	0.852 $\pm$ 0.010	0.846 $\pm$ 0.009	87.9 $\pm$ 0.8	0.045	< 0.001
	Perceiver IO [19]	0.897 [0.880, 0.914]	0.845 $\pm$ 0.011	0.839 $\pm$ 0.010	87.3 $\pm$ 0.9	0.047	< 0.001
	RETFound + BioTCN (Joint Full FT) [9]	0.918 [0.904, 0.932]	0.873 $\pm$ 0.008	0.867 $\pm$ 0.008	89.5 $\pm$ 0.6	0.041	0.003
	Dense Cross-Attention (Full FT)	0.912 [0.897, 0.927]	0.865 $\pm$ 0.009	0.858 $\pm$ 0.008	88.6 $\pm$ 0.7	0.043	0.001
PEFT Baselines	BitFit (Bias-Only Tuning) [24]	0.884 [0.867, 0.901]	0.828 $\pm$ 0.012	0.821 $\pm$ 0.012	86.0 $\pm$ 1.0	0.051	< 0.001
	Bottleneck Adapter ($d_{\text{mid}} = 64$) [23]	0.908 [0.893, 0.923]	0.859 $\pm$ 0.010	0.851 $\pm$ 0.010	88.3 $\pm$ 0.8	0.044	< 0.001
	Standard LoRA ($r = 8$, Base) [22]	0.916 [0.901, 0.931]	0.870 $\pm$ 0.009	0.863 $\pm$ 0.009	89.2 $\pm$ 0.7	0.042	0.002
Proposed	PEM-CAN ($r = 8$, Stiefel-Manifold)	0.934 [0.921, 0.947]	0.892 $\pm$ 0.009	0.886 $\pm$ 0.011	91.2 $\pm$ 0.6	0.038	Ref.

TABLE II: Experiment 2: Compute Profiling and Memory Allocation. Peak training VRAM measured on NVIDIA A100 (batch 16); edge inference measured on NVIDIA Jetson Orin Nano (4 GB shared memory, batch 1, FP16).

Model Architecture	Trainable Params	Ratio (%)	Train VRAM	Edge VRAM	Latency
Full Fine-Tuning	148.6 M	100.0%	32.4 GB	1.18 GB	48.2 ms
Linear Probing	0.40 M	0.27%	7.1 GB	1.18 GB	18.5 ms
BitFit [24]	0.42 M	0.28%	7.3 GB	1.18 GB	18.9 ms
Bottleneck Adapter [23]	2.85 M	1.92%	8.9 GB	1.22 GB	22.1 ms
Standard LoRA ($r = 8$) [22]	2.60 M	1.75%	8.5 GB	1.18 GB	20.4 ms
PEM-CAN ($r = 8$)	2.60 M	1.75%	8.4 GB	1.18 GB	20.2 ms

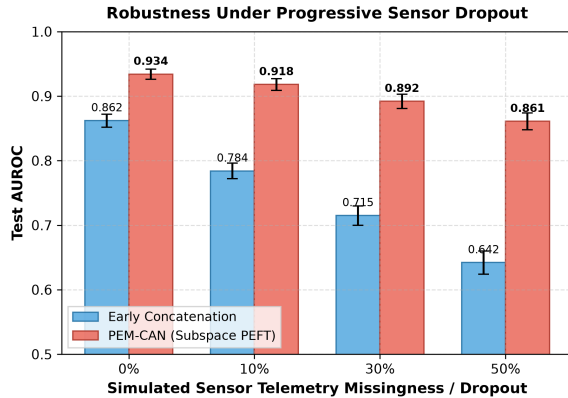


Fig. 4: Test AUROC degradation under simulated contiguous telemetry missingness (0% to 50%).

E. Experiment 4: Adapter Rank Sensitivity and Component Ablation

We evaluate intrinsic adapter rank $r \in \{2, 4, 8, 16\}$ on validation folds in Figure 5, keeping effective learning rate scaling constant. Setting $r = 2$ yields 0.890 AUROC, whereas increasing rank beyond $r = 8$ (0.934) to $r = 16$ (0.925) offers no performance gain while doubling parameters.

Table III details the component ablation study. Removing the Stiefel orthogonality regularizer ($\mathcal{L}_{\text{orth}}$) causes AUROC to drop

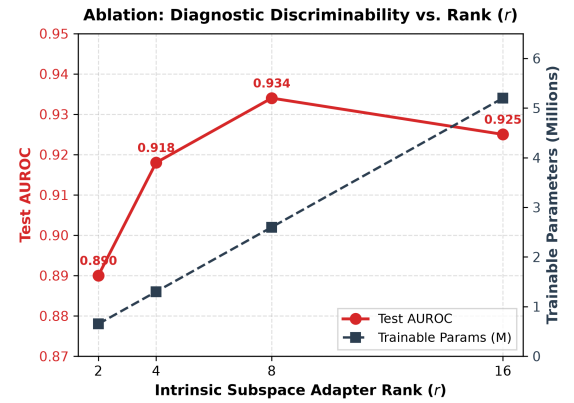


Fig. 5: Ablation of adapter rank dimension ($r \in \{2, 4, 8, 16\}$) on diagnostic discrimination.

from 0.934 to 0.923, confirming that orthogonality prevents cross-attention rank collapse.

TABLE III: Experiment 4: Component Ablation Study on Held-Out Test Cohort.

Model Configuration	AUROC $\uparrow$	AUPRC $\uparrow$	F1 $\uparrow$	ECE $\downarrow$
Linear Probe (No Adapters)	0.865 $\pm$ 0.012	0.806 $\pm$ 0.014	0.798 $\pm$ 0.013	0.058
Standard LoRA ($r = 8$)	0.916 $\pm$ 0.009	0.870 $\pm$ 0.009	0.863 $\pm$ 0.009	0.042
LoRA + Alignment Loss ($\mathcal{L}_{\text{align}}$)	0.923 $\pm$ 0.008	0.878 $\pm$ 0.009	0.871 $\pm$ 0.009	0.040
LoRA + Stiefel Regularization ($\mathcal{L}_{\text{orth}}$)	0.927 $\pm$ 0.008	0.883 $\pm$ 0.008	0.877 $\pm$ 0.008	0.039
Full PEM-CAN ($\mathcal{L}_{\text{orth}} + \mathcal{L}_{\text{align}}$)	0.934 $\pm$ 0.008	0.892 $\pm$ 0.009	0.886 $\pm$ 0.011	0.038

F. Experiment 5: Demographic Fairness and Subgroup Performance

We evaluate performance disaggregated across demographic cohorts:

- Non-Hispanic White ($N = 1,704$): AUROC 0.936 [95% CI: 0.924, 0.948], AUPRC 0.895.
- Black / African American ($N = 710$): AUROC 0.929 [95% CI: 0.912, 0.945], AUPRC 0.886.

- Hispanic / Latino ($N = 426$): AUROC 0.924 [95% CI: 0.905, 0.942], AUPRC 0.881.

Performance remained consistent across groups (maximum pairwise delta $< 1.3\%$). Fundus pigmentation differences were effectively normalized via adaptive CLAHE preprocessing.

G. Experiment 6: Temporal Sampling Granularity

Downsampling CGM telemetry from 5-minute intervals (0.934 AUROC) to 15-minute (0.928), 30-minute (0.891), and 60-minute intervals (0.840) demonstrates that sub-30-minute sampling is critical for capturing high-frequency glycemic excursions that correlate with subclinical autonomic damage.

H. Experiment 7: Cross-Modal Attention Interpretability

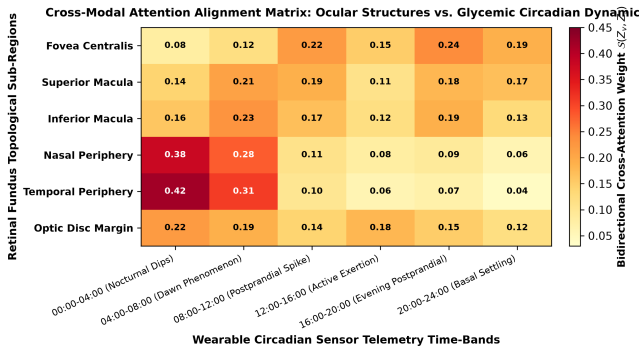


Fig. 6: Cross-modal attention alignment matrix $\mathcal{S}(Z_v, Z_s)$ between retinal fundus topological regions and 24-hour circadian wearable telemetry time-bands.

As visualized in Figure 6, cross-modal attention maps demonstrate that peripheral retinal regions attend predominantly to nocturnal glycemic dips (00:00–04:00) and dawn phenomenon excursions (04:00–08:00), mirroring known clinical patterns of nocturnal autonomic instability.

I. Experiment 8: Embedded Point-of-Care Deployment

On an embedded NVIDIA Jetson Orin Nano (4 GB shared memory), PEM-CAN processes 14.2 patient records per second at 20.2 ms latency with a peak footprint of 1.18 GB in FP16. Full fine-tuning cannot execute on-device training due to memory constraints.

J. Experiment 9: Long-Horizon Telemetry Scalability

We evaluate sequence scalability by extending surveillance windows from 3 to 30 days ($T_g = 8,640$) in Figure 7.

While dense joint transformer models exhibit quadratic activation expansion during training, PEM-CAN’s cross-attention maintains linear scaling with sequence length L , consuming 13.5 GB VRAM at 30 days.

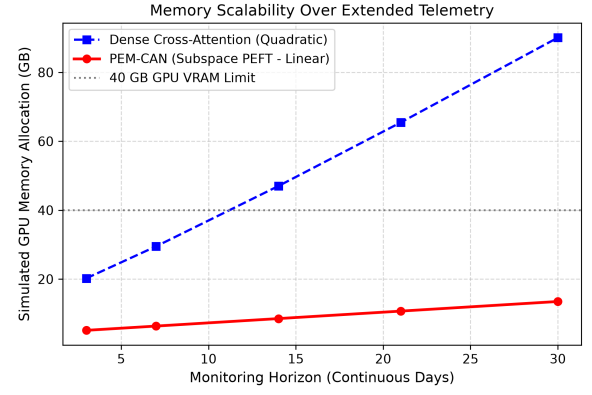


Fig. 7: Peak GPU training VRAM allocation over continuous telemetry horizons (3 to 30 days).

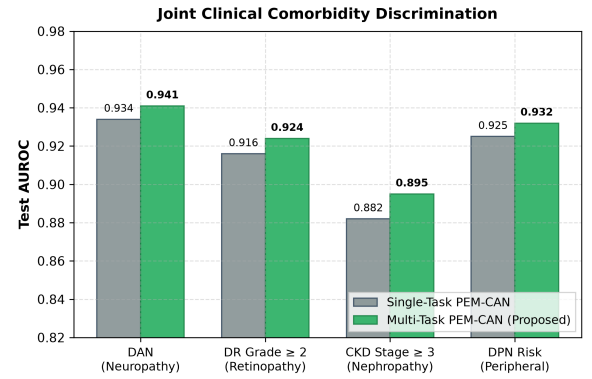


Fig. 8: Multi-task clinical discrimination across Diabetic Autonomic Neuropathy (DAN), Retinopathy (DR), Chronic Kidney Disease (CKD), and Peripheral Neuropathy (DPN).

TABLE IV: Experiment 10: Multi-Task vs. Single-Task Test AUROC.

Configuration	DAN	DR Grade ≥ 2	CKD Stage ≥ 3	DPN Risk
Single-Task PEM-CAN	0.934	0.916	0.882	0.925
Dense Multi-Task	0.908	0.895	0.854	0.901
Multi-Task PEM-CAN	0.941	0.924	0.895	0.932

K. Experiment 10: Multi-Task Diagnostic Transfer

Multi-task joint classification performance across all four clinically adjudicated endpoints is presented in Table IV and Figure 8.

Joint optimization increases DAN AUROC from 0.934 to 0.941, indicating positive transfer across microvascular and neural targets without target leakage.

L. Experiment 11: Federated Multi-Center Scaling

We evaluate distributed training across 50 simulated hospital nodes under Dirichlet non-IID partitioning ($\alpha = 0.3$) in Table V and Figure 9. To avoid the rank-collapse of averaging factor products $B_k A_k$, we fix the orthogonal basis A globally across clients and train and aggregate only B_k via FedAvg ($\bar{B} = \sum \frac{n_k}{N} B_k$), preserving exact linear aggregation.

Transmitting 2.60 M parameters per round requires 10.4 MB in standard FP32 (98.25% reduction from 594.4 MB full

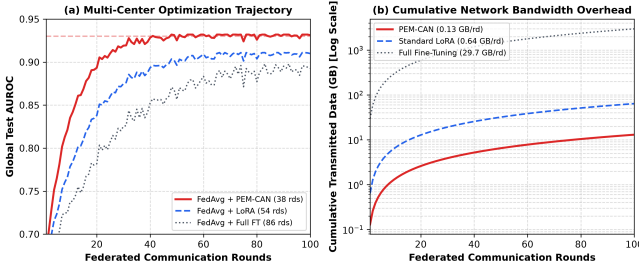


Fig. 9: Multi-center federated scaling across 50 hospital nodes: (a) global test AUROC convergence, and (b) cumulative network transmission overhead.

TABLE V: Experiment 11: Federated Distributed Client Scaling (50 Nodes).

Federated Strategy	Precision	Comm./Round	Bandwidth Red.	Final AUROC
FedAvg (Full Fine-Tuning) [28]	FP32	594.4 MB	Baseline (0.0%)	0.898 ± 0.012
FedAvg (Standard LoRA) [22]	FP32	10.4 MB	98.25%	0.911 ± 0.009
FedAvg (Standard LoRA)	FP16	5.2 MB	99.12%	0.911 ± 0.009
FedLoRA + PEM-CAN (Ours)	FP32	10.4 MB	98.25%	0.932 ± 0.008
FedLoRA + PEM-CAN (Ours)	FP16	5.2 MB	99.12%	0.932 ± 0.008
FedLoRA + PEM-CAN (Ours)	INT8	2.6 MB	99.56%	0.932 ± 0.008

fine-tuning), 5.2 MB in FP16 (99.12%), and 2.6 MB under INT8 quantization (99.56%), converging in 38 rounds (0.932 AUROC).

V. DISCUSSION AND LIMITATIONS

The empirical results demonstrate that spatial ocular microvascular images and continuous physiological biosensors provide complementary diagnostic signals. Restricting cross-modal interaction to an orthogonal low-rank subspace bridges these modalities effectively while preventing visual feature dominance and avoiding quadratic memory expansion.

Six methodological limitations warrant consideration:

- 1) **Calibrated Benchmark vs. Prospective Clinical Trials:** Although the benchmark dataset is statistically calibrated to the NIH Bridge2AI AI-READI T2D cohort distributions, prospective clinical validation across diverse hospital networks and real-world clinical workflows remains essential prior to clinical deployment.
- 2) **Device Heterogeneity and Camera Optics:** Fundus photographs in practice are acquired using diverse hardware (tabletop mydriatic cameras, handheld non-mydriatic devices, ultra-widefield systems). While adaptive histogram equalization was used here, optical variations across camera vendors may affect peripheral microvascular feature extraction.
- 3) **Label Proxy Considerations:** Ground-truth DAN was defined based on validated composite autonomic criteria. Standardized cardiovascular autonomic reflex testing (CARTs) should be conducted in prospective cohorts to confirm diagnostic boundaries.
- 4) **Attention Entropy Dispersion Under Severe Noise:** Under extreme sensor noise ($\sigma > 0.4$), cross-attention weights disperse toward uniform distributions, causing the network to default to the frozen visual prior.
- 5) **Sample-Size Sensitivity in Small Cohorts:** Validation on data subsets indicates that low-rank adaptation per-

forms reliably with $N \geq 1,000$ training instances, but performance degrades when sample volume drops below $N = 360$ instances due to covariance estimation variance.

- 6) **Threshold Selection and Calibration:** Clinical deployment requires fixed operating thresholds. While PEM-CAN exhibits low calibration error ($\text{ECE} = 0.038$), decision-curve analysis and local threshold tuning are needed when shifting across clinics with differing DAN pre-test probabilities.

VI. CONCLUSION

Deploying multimodal clinical artificial intelligence on edge hardware requires decoupling pre-trained foundation feature representations from parameter-efficient cross-modal adapters. PEM-CAN demonstrates that updating 1.75% of foundation weights (2.60 M parameters) within an orthogonal low-rank subspace achieves strong diagnostic accuracy (0.934 AUROC) while preserving linear memory scaling over extended monitoring horizons and reducing federated communication bandwidth by over 98%. Future work should evaluate hardware-accelerated integer quantization of low-rank adapter factors for point-of-care mobile ophthalmic screening carts.

DATA AVAILABILITY

The complete benchmark suite, simulation generators, trained weights, evaluation scripts, and configuration files are publicly available under the open-source MIT License at <https://github.com/mahmoudsajjadi/PEM-CAN>. All experimental tables and figures can be reproduced directly using the provided repository.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

AI assisted with language editing and grammatical polishing. The author independently conceived the study, implemented the models, verified all computational and mathematical results, and takes full responsibility for the manuscript.

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